

Best-responding to whom? Risk attitudes and opponent-conditioning of LLM agents in collective-risk dilemmas

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Abstract

Language-model agents are being deployed into settings where their choices impose risk on a group, yet most of the evidence about how they behave there comes from self-play, in which one model faces copies of itself. Self-play cannot separate an agent reasoning about its opponents from an agent executing a fixed script, because the two produce the same transcript. We run 1850 games of the collective-risk social dilemma across five production models and break that symmetry three ways. On an eleven-point grid of catastrophe risk, three of five models contribute the same amount at every level, and a fourth changes only at the control at $p = 0$. There contributing is strictly dominated, yet 240 of 300 seats still pay, and the panel forfeits 46% of the payoff attainable at the welfare-optimal outcome. Deleting the prompt’s equal-split gloss never lowers contribution and raises it to the endowment cap in two models, so the cooperation is not compliance with a supplied focal point. Nor is it a failure of comprehension: asked in the middle of their own games, the agents answer 3,600 questions about the rules at 100.0% and identify the higher-paying strategy at $p = 0.1$ at 94.0%, then contribute 22.2 units against their own stated answer. Finally, seating one model against 5 scripted opponents makes the best response computable: no model plays it in any of 1000 games, including 500 in which every alternative is strictly dominated, and a trivial free-rider out-earns every model in the panel. Cooperation from these agents is a fixed schedule rather than a choice, which makes them safe collaborators and unsafe stakeholders.

CCS Concepts

• Computing methodologies → Multi-agent systems; Natural language processing.

Keywords

Large language model agents; social dilemmas; collective risk; best response; empirical game theory

1 Introduction

A language-model agent that manages a budget, a shared queue or a resource allocation is a player in a social dilemma whether or not its designer framed it as one. Its choices carry a cost to itself and a benefit to a group, and the group may fail. What a designer needs to know before deploying such an agent is narrow and practical: does it defend the resources it was given, and does what it does depend on what the other agents do? An agent that contributes whatever the situation is not a cooperator but a liability, since a single opportunistic counterpart converts its generosity into a transfer.

Answering either question is exactly what the standard experimental design cannot do. Studies of language-model agents in social dilemmas commonly seat copies of a single model at every position [9, 11], and where the design does vary the opponent it

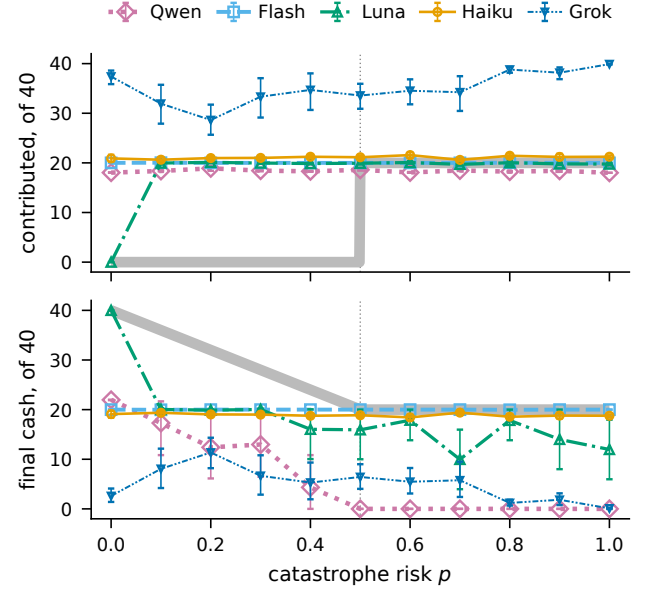


Figure 1: Self-play across the whole range of catastrophe risk, 550 games, 10 per cell, with percentile bootstrap 95% intervals. The thick grey line is the optimum of Proposition 1, which is simultaneously the welfare-maximising outcome and the best Nash equilibrium: contribute nothing below $p^* = 1/2$ (dotted vertical), contribute exactly the share of 20 above it. It is drawn beneath the model lines so that a model sitting on it stays visible. Top, what a seat contributes of its 40-unit endowment; bottom, what it finishes with. No line bends at the step. The two panels together separate the failure modes: GROK-4.20-NON-REASONING above the optimum and QWEN3-235B-A22B below it arrive at the same floor in the lower panel, one by buying a target it did not need and one by paying for a target it never reaches.

is usually a two-player repeated game [1, 4]. Self-play with more than two seats produces a transcript that a model reasoning about its opponents and a model executing a fixed contribution schedule generate alike, so the observation “the agents cooperated” is compatible with both, and only one of them survives contact with an opponent who defects. The prompt compounds the problem: a dilemma stated with a group target and a round count states its own equal split, and an agent that repeats that split scores as a cooperator on every aggregate measure.

We break the symmetry of self-play in a collective-risk social dilemma [8], where six players contribute from private endowments towards a group target and a catastrophe lottery fires with probability p if they miss it. The game suits the question because p

is a dial the experimenter owns. It moves the optimum from “contribute nothing” to “contribute exactly your share” at a threshold we derive in closed form, so an agent that is choosing and an agent that is repeating are told apart by whether anything changes when the optimum does. We run 1850 games on five production models — one per provider, all at the cheap tier of their family — and apply three controls, each of which removes one explanation for the cooperation the models display.

The first control varies the risk itself over an eleven-point grid whose endpoint, $p = 0$, removes the catastrophe altogether and thereby makes contributing strictly dominated: no belief and no attitude to risk can justify a single unit. The second deletes the two spans of the prompt that state the equal split, which tests whether the models are solving the dilemma or reciting the instructions. The third seats one model against 5 opponents whose policies we fix in advance, which makes the best response an arithmetic fact rather than an estimate. Alongside all three we ask the playing agent, in the middle of its own game, what the rules are and which of two named strategies pays more, so that a shortfall in play can be attributed to knowledge or denied it.

Our contributions are the following.

- **A risk-response measurement with a control that needs no assumptions (§4).** We characterise the equilibria of the game as a function of p and show that welfare-optimal and equilibrium play switch at the same threshold, $p^* = 1/2$. Against that benchmark three of five models contribute a constant across all 11 risk levels, the fourth changes only at $p = 0$, and 240 of 300 seats pay at $p = 0$, where paying is strictly dominated. The panel forfeits 46% of attainable payoff (Figure 1).
- **Two controls that rule out the cheap explanations (§5).** Deleting the prompt’s equal-split gloss never lowers contribution and raises it to the endowment cap in two models, so the behaviour is not compliance with a supplied focal point. The agents answer 3,600 in-situ questions about the rules at 100.0% and, at $p = 0.1$, identify contributing nothing as the higher-paying strategy at 94.0% — then contribute 22.2 units in the same games. Knowing the answer and acting on it come apart.
- **A best-response profile against known opponents (§6).** With 5 scripted seats the best response is computable exactly. No model plays it in any of 1000 games, including the 500 in which contributing is strictly dominated, and a free-rider that contributes nothing out-earns every model in the panel.

Read together, the three controls say the same thing from three directions: the cooperation these agents display is a schedule rather than a choice. A schedule makes them safe collaborators and unsafe stakeholders, and that distinction is one a deployment can design around only once it is measured.

2 Related work

Collective-risk dilemmas. The game we use was introduced to study why groups fail to avert a shared catastrophe [8]: six players contribute from private endowments towards a threshold, and missing it triggers a loss with a probability the experimenter sets.

Human groups reach the target reliably only when the probability is high, and inequality between players and the ability to communicate both shift the outcome [13]. Evolutionary treatments explain the pattern by the same threshold structure we exploit here: risk of collective loss turns a defection equilibrium into a coordination problem with multiple equilibria, and which one a population reaches depends on the risk [12]. That literature establishes the game as a place where behaviour *should* move with p , which is what makes a flat response diagnostic rather than merely disappointing. Our contribution is not a new account of human play but a measurement instrument applied to artificial players.

Social dilemmas in multiagent systems. Sequential social dilemmas gave multiagent reinforcement learning a setting in which cooperation is a policy property rather than a matrix entry [7, 10], and subsequent work showed that changing the learner’s objective — inequity aversion, for instance — changes which equilibrium a population settles into [6]. The programme of cooperative AI states the design question directly: build agents that can cooperate with agents unlike themselves [3]. Language-model agents enter that programme with a property no learned policy has, in that their behaviour is fixed at deployment rather than shaped by the population they meet, and the present paper measures what that fixed behaviour is.

Language-model agents in games. Language models have been placed in canonical games and read as economic subjects [5], played against each other and against scripted strategies in two-player repeated games [1, 4], and embedded in societies that manage a shared resource [11] or an open-ended social world [9]. Two design choices recur. Beyond two players the seats usually hold copies of one model, which leaves the agent’s response to a *different* policy unmeasured; and the reported quantity is usually an outcome, such as whether the resource survived, rather than a distance from a computed optimum. We keep the game and change both choices: opponents are fixed policies chosen so that the best response can be enumerated, and every result is reported against that best response.

Overlap with concurrent work. A concurrent manuscript uses the same game and an overlapping panel to ask whether language models reproduce the risk sensitivity of human subjects, and reports the single-model baseline that Section 4 re-runs and extends [2]. The questions differ: that work asks whether a model is risk-sensitive, and treats the answer as a statement about machine behaviour; the present paper asks what a model does when the other agents are not copies of itself, and treats the answer as a statement about deploying agent populations. The results in Sections 5 and 6, which carry this paper, have no counterpart there.

3 The game, its equilibria, and the panel

3.1 The game

The six players each hold an endowment of $e = 40$ units. Over $R = 10$ rounds every player privately chooses a contribution $a_i^t \in \{0, 2, 4\}$, and after each round every player sees what all six chose in every round so far. Writing $x_i = \sum_t a_i^t$ for a player’s total and $X = \sum_i x_i$ for the group’s, the group succeeds if $X \geq T = 120$. A

player’s cash is then what it kept:

$$u_i(x) = \begin{cases} e - x_i, & X \geq T, \\ 0 \text{ w.p. } p, \quad e - x_i \text{ w.p. } 1 - p, & X < T. \end{cases} \quad (1)$$

Contributions are never refunded, the round count is stated in the prompt, and $R \cdot \max A = e$, so no player can contribute more than it holds. The risk p is the only quantity we vary in Section 4. Since (1) depends on the action profile only through the totals x , the Nash equilibrium *outcomes* of the R -round game are those of the reduced game in which each player picks $x_i \in \{0, 2, \dots, e\}$, and we characterise them there.

Proposition 1. *Fix $p < 1$. A profile x is a Nash equilibrium if and only if either $x = 0$, or $\sum_i x_i = T$ and $x_i \leq p e$ for every i . Hence an equilibrium that reaches the target exists if and only if $p \geq p^*$ with $p^* = T/(Ne) = 1/2$, and at $p = p^*$ the only such equilibrium is the equal split $x_i = T/N = 20$. The welfare-maximising outcome is $x = 0$ for $p < p^*$ and $\sum_i x_i = T$ for $p > p^*$, so the per-capita payoff of the best outcome is $\max\{(1-p)e, T/N\}$.*

Proof sketch. If $X < T$ and $x_i > 0$, then moving to $x_i = 0$ leaves the target missed and raises the expectation from $(1-p)(e-x_i)$ to $(1-p)e$, strictly for $p < 1$; so an equilibrium that misses the target has $x = 0$, and $x = 0$ is one because a lone player can contribute at most $e < T$. If $X > T$ then $X \geq T + 2$, and any i with $x_i \geq 2$ sheds 2, still reaches the target and gains 2; so an equilibrium that reaches it has $X = T$ exactly. Given $X = T$, every profitable-looking deviation reduces x_i and therefore misses the target, and the best of them is $x_i = 0$, worth $(1-p)e$ against $e - x_i$ for staying; the two compare as $x_i \leq p e$. Summing that bound over N players against $\sum_i x_i = T$ gives $p \geq T/(Ne)$, with equality forcing $x_i = T/N$ for all i . For welfare, $\sum_i u_i$ equals $Ne - X$ when $X \geq T$ and $(1-p)(Ne - X)$ otherwise, so it is maximised at $X = T$ or at $X = 0$, and $Ne - T \geq (1-p)Ne$ reduces to $p \geq T/(Ne)$ again. ■

Three consequences carry the rest of the paper. Welfare-optimal play and best-equilibrium play switch at the *same* threshold, so the benchmark $\text{opt}(p) = \max\{(1-p)e, T/N\}$ used throughout needs no choice between the two. Below p^* no equilibrium reaches the target, which is to say that a group that reaches it there is not merely being generous but is playing an outcome no player should sustain. And at $p = 0$ the two branches of (1) coincide, so $u_i = e - x_i$ whatever the group does and contributing is strictly dominated — under any increasing utility, not only under the risk-neutral one that opt assumes. The $p = 0$ column is therefore the one place in the paper where no assumption about risk attitude is doing any work, and we lean on it accordingly.

3.2 Panel and protocol

We run five production models, one per provider and each at the inexpensive tier of its family, so that no result is an artefact of a single vendor’s post-training: CLAUDE-HAIKU-4.5 (Haiku), GEMINI-3.5-FLASH-LITE (Flash-Lite), GPT-5.6-LUNA (Luna), QWEN3-235B-A22B-INSTRUCT (Qwen) and GROK-4.20-NON-REASONING (Grok), named by their short forms from here on. Every experiment runs every model on exactly the same number of games; price never determines sample size. All games are in English, at temperature 0.7, with every seat holding a neutral persona.

model	$p = 0$		all 11 risk levels				slope in p	
	giv.	cash	giv.	hit%	cash	lost%	$\hat{\beta}$	P
Qwen3-235B	18.0	22.0	18.3	2	6.3	75	-0.20	0.104
Flash-Lite 3.5	20.0	20.0	20.0	100	20.0	21	0.00	0.911
GPT-5.6 Luna	0.0	40.0	18.1	70	18.5	27	8.88	<0.001
Haiku 4.5	20.9	19.1	21.1	98	18.9	26	0.44	0.064
Grok-4.20 NR	37.4	2.6	35.0	100	5.0	80	6.32	<0.001

Table 1: Self-play on the eleven-point risk grid, 550 games. “giv.” is the mean total a seat contributes of its 40-unit endowment, “cash” its mean final payoff, “hit%” the share of games reaching the target. “lost%” is the share of the attainable payoff forfeited, measured against $\max\{(1-p)40, 20\}$, which is both the welfare-optimal outcome and the best Nash equilibrium (Proposition 1). The two $p = 0$ columns need no such benchmark: there contributing is strictly dominated, so every unit given is lost under any increasing utility. $\hat{\beta}$ is the OLS slope of “giv.” on p , with P from a 10,000-resample permutation test over risk labels; the unit of analysis is the game.

Each round a seat receives the rules, the complete per-seat history in fixed positions, and a request for one contribution. The running group total is *not* printed, so a seat that wants it must add the history up; we keep it that way because printing it changes behaviour, and because Section 6 needs a measure that uses only what the prompt displays. A reply is parsed from a CONTRIBUTION: line. Replies that reach the output cap before that line are retried at a quadrupled cap rather than parsed, because a truncated chain of arithmetic otherwise yields a plausible number that was never a decision; every game reported here has zero parse failures and zero truncations across 111,000 decisions.

Cells hold 10 games each, indexed by a repetition number that also seeds the catastrophe lottery, so the same repetition faces the same draw at every risk level — common random numbers, which is what makes the paired comparisons across p in Section 4 worth making. The unit of analysis is the game, never the seat: six seats in one game read each other’s history and are not independent. Every interval is a percentile bootstrap over games with 10,000 resamples and every P -value a permutation test with 10,000 resamples, both under a fixed seed.

4 Cooperation that does not depend on the risk

We first run every model in self-play across the full range of the one parameter that should matter. Six seats of one model play the game of Section 3 at each of eleven catastrophe probabilities from 0 to 1 in steps of 0.1, 10 repetitions per cell, 550 games in all. Proposition 1 says what should change along that axis: below $p^* = 1/2$ no equilibrium reaches the target and the best outcome is to contribute nothing, above it the equal split of 20 each becomes both an equilibrium and the welfare optimum. Figure 1 plots what the models do against that step, and Table 1 gives the numbers.

The schedule does not move when the optimum does. Of five models, three show no detectable dependence on p at all (Qwen, Flash-Lite and Haiku; largest slope 0.44 units per unit of p , $P = 0.064$). One of them is flat to the resolution of the instrument: Flash-Lite

contributes 20.0 units with a standard deviation of 0.04 across all 110 of its games, and plays exactly 2 in every round of 99.7% of seats. That constant is the equilibrium of $p = p^*$ and of no other risk level, so the model plays the knife-edge case everywhere, including the ten levels at which it is not an equilibrium at all. The two models whose slope is non-zero do not rescue the panel. Luna appears to respond (8.88, $P < 0.001$), but the entire response is one switch: it contributes 0.0 at $p = 0$ and 19.9 on average everywhere else, and deleting the $p = 0$ column leaves a slope of -0.24 that a permutation test cannot separate from zero ($P = 0.108$). It detects that a risk of exactly zero is different; it does not grade. Grok does climb over $p > 0$ (10.02, $P < 0.001$), but never from a level any equilibrium supports: it ranges from 28.6 units at $p = 0.2$ to 39.9 at $p = 1$. Nor is the climb monotone. At $p = 0$ it contributes 37.4 units, more than at 7 of the 10 positive risk levels — and at $p = 0$ the parameter it is apparently tracking has vanished.

The $p = 0$ control, which assumes nothing. Everything above prices behaviour against a risk-neutral benchmark, and a reader is entitled to answer that these agents may simply be risk-averse. At $p = 0$ that answer is unavailable. With no catastrophe the two branches of (1) coincide, a player keeps $40 - x_i$ whatever the group does, and contributing is strictly dominated under every increasing utility. Of the 300 seats played there, 240 pay anyway — 80% — and 40 of 50 games see money spent on a threshold that protects against nothing. Two models reach the target in every single game at $p = 0$ (Flash-Lite and Grok) and a third in 80%. Only Luna contributes nothing, and it collects the full 40-unit endowment for doing so, against 2.6 units for Grok in the same cell. The gap between those two numbers is not a difference in risk attitude, because there is no risk.

What it costs. Proposition 1 makes the accounting unambiguous, since the welfare-optimal outcome and the best equilibrium pay the same $\text{opt}(p) = \max\{(1 - p)40, 20\}$ and no choice between them is needed. Measured against it the panel takes home 13.7 units of an available 25.5, forfeiting 46%. The loss is not evenly spread: 21% for Flash-Lite, 26% for Haiku and 27% for Luna, against 75% for Qwen and 80% for Grok. A deployment that reads only the success rate would rank the panel in almost the opposite order, because the two models that lose the most are one that always reaches the target and one that almost never does.

Two ways to lose, and success rate hides both. Grok reaches the target in 100% of games and suffers no catastrophe in any of them, which is what a group-level metric rewards; it achieves this by contributing 35.0 of its 40 units, so its seats finish with 5.0. Qwen fails from the other side, missing the target in 108 of its games while still paying 18.3 units into a pot that never reaches 120; the lottery then takes the remainder in 78 games and its seats finish with 6.3. One model buys a certainty it did not need and the other buys nothing at all, and both end near zero. Only a benchmark that prices the outcome against what was attainable puts them side by side, which is why we report the gap rather than the success rate throughout.

model	focal point deleted (§5.1)				asked outright (§5.2)			
	with	w/o	Δ	P	rules	EV	EV _{.1}	giv _{.1}
Qwen3-235B	18.5	34.8	16.4	<0.001	100	33	100	18.0
Flash-Lite 3.5	20.0	20.1	0.0	0.294	100	99	100	20.0
GPT-5.6 Luna	19.9	19.8	-0.1	0.490	100	84	93	20.0
Haiku 4.5	21.0	21.6	0.6	0.012	100	100	100	20.2
Grok-4.20 NR	34.6	40.0	5.4	<0.001	100	46	77	32.8

Table 2: Two controls, on the same panel. Left: mean total contributed per seat with the prompt’s equal-split gloss present (“with”) and deleted (“w/o”), at $p \in \{0.1, 0.5, 0.9\}$, 30 games each; P from a 10,000-resample permutation test on games. Deleting the focal point never lowers contribution. Right: accuracy (%) on in-situ questions asked of the playing agent — “rules” over 3,600 questions on the mechanics, “EV” on the expected-value comparison between contributing one’s share and contributing nothing. “EV_{.1}” restricts that comparison to $p = 0.1$, where the correct answer is that contributing nothing pays more, and “giv_{.1}” is what the same seats contributed in the same games. Read the EV columns together with §5.2: Qwen and Grok give the same answer at every risk level, so their EV_{.1} scores are a coincidence rather than a computation, and the dissociation is claimed only for the three models that answer differently at each level and get each one right.

5 Two things the cooperation is not

Section 4 leaves two cheap explanations standing. The models may be repeating a solution the prompt handed them, since a dilemma stated with a target, a group size and a round count also states its own equal split; or they may not understand the game well enough for the risk to mean anything. We test both on the same panel, at $p \in \{0.1, 0.5, 0.9\}$, and Table 2 reports them side by side.

5.1 Deleting the focal point

The prompt states the equal split twice: once as a gloss on the target (“an average of 2 per player per round”) and once as a worked example on the payoff line (“contributing 2 every round leaves you 20 at the end”). We delete both spans and change nothing else, which leaves the target, the group size and the round count in place, so the split remains derivable by one division. Each model plays 30 games with the spans deleted against the 30 baseline games at the same risk levels.

Contribution never falls. Three models rise significantly (Qwen, Haiku and Grok) and two do not move at all (Flash-Lite and Luna); none contributes less. The two largest moves are the ones that matter, because both go the wrong way for the instruction-following account. Qwen goes from 18.5 units to 34.8 ($P < 0.001$) and its target rate from 0% to 100%, yet its final cash falls from 5.8 to 5.2: it reaches the goal by spending most of its endowment on it. Grok goes from 34.6 to 40.0 of 40 ($P < 0.001$), the share of its seats contributing the maximum in every round rising from 50.6% to 98.9%, and its cash falls to 0.0. Removing the sentence that names a moderate contribution does not remove the cooperation; it removes the ceiling on it.

The models that do not move show where the split actually comes from. Flash-Lite plays exactly 2 in every round of 97.2% of its seats without the gloss, against 99.4% with it, and Luna moves by -0.1 units ($P = 0.490$). They reconstruct the split rather than reciting it, which they can: 120 over six players over ten rounds is one division. The equal split is therefore a focal point of the *game*, not a phrase in the prompt, and deleting the phrase leaves the behaviour it was supposed to explain intact while making two models worse off.

5.2 Asking the agent what it knows

Between rounds we put questions to the same agent that is playing, in a separate call whose answer never enters the game history, so a probe cannot steer the play it measures. Each game carries 30 questions at rounds 1, 5, 10, giving 4500 in total: 3,600 on the mechanics — the endowment, the target, the legal contributions, the round count, the stated risk, the payoff after a disaster — and the rest on value. The value question we rely on names both strategies in words and asks which pays more on average: contributing 2 every round and finishing with a certain 20, or contributing nothing and keeping 40 unless the disaster strikes.

Comprehension is not the obstacle. The panel answers all 3,600 rule questions correctly, 100.0%, with no model below it. On the value comparison the panel scores 72.4% overall and 94.0% at $p = 0.1$, where the right answer is that contributing nothing pays more — 36 against 20. The aggregate hides a distinction the same data settles. Qwen and Grok return the same answer at every risk level (Qwen gives one value in 100% of its answers), so their score at $p = 0.1$ is a coincidence and not a computation. The other three models change their answer with p and are right each time: 100% for Haiku across all three levels, 99% for Flash-Lite and 84% for Luna.

Those three then play against their own answer. In the very games in which they state that contributing nothing pays more, they contribute 20.2, 20.0 and 20.0 units respectively, which is the equal split to within a rounding error in all three cases. The comparison is within-agent and within-game rather than across conditions, so no difference in prompt, sampling or model version can absorb it. Whatever produces the contribution, it is not reached by evaluating the two strategies the agent has just evaluated correctly.

6 Best-response profiling against known opponents

Every result so far comes from one model playing copies of itself, a design that cannot separate a model reasoning about its opponents from a model following a script. We remove the symmetry: a single LLM is seated against five opponents whose policies we fix in advance. Because the opponents are then known, the best response can be computed exactly, and “does the model best-respond?” becomes arithmetic.

6.1 Design

Seat 1 is an LLM; the other 5 run one deterministic policy, identical across seats and constant within a game — ALL-DEFECT contributes 0 every round, ALL-CONTRIBUTE 4, FAIR-SHARE 2 (the per-round equal split of the target, which the prompt names), and COND. COOP.

the round-average of the other five seats, rounded to a legal amount. Crossing these with 5 risks $p \in \{0.1, 0.3, 0.5, 0.7, 0.9\}$, 5 models and 10 repetitions gives 100 cells and 1000 games; the game is otherwise that of Section 3. Only one seat calls a model, so a game costs 10 API calls rather than 60.

Two protocol facts govern every reading below. The LLM is *not* told that its opponents are deterministic, scripted or identical; it receives the prompt of the symmetric game, in which the pool is never printed but the complete per-seat history is shown each round in fixed positions, so the pool is recoverable by addition. That history is what keeps the omission from being decisive: under ALL-DEFECT and ALL-CONTRIBUTE the five scripted seats return the same five numbers every round, so a player entering round 6 has seen 5 identical rounds and the opponents’ policy is an induction from the transcript rather than a fact we withheld — and one the models demonstrably perform (Figure 2). We therefore separate round 1, where no history exists and nothing can be inferred, from rounds 6–10, and rest the claim on the latter. And before computing anything we replayed all five scripted seats of all 1000 games from the LLM’s recorded moves, requiring a round-by-round match; all replay exactly, which is what licenses treating the opponents as known.

6.2 What a rational seat would do

With deterministic opponents the LLM faces a decision problem, solved by exhaustive search over all 3^{10} trajectories under the game’s own scoring rule.

Proposition 2. *Against ALL-DEFECT and ALL-CONTRIBUTE the best response is to contribute 0 at every p . Against FAIR-SHARE and COND. COOP. it is $0 \rightarrow 20$, switching at $p^* = 0.5$, where both are optimal.*

Proof sketch. The scripted seats bank 0, 200 and 100 in the first three profiles. Against ALL-DEFECT the group total cannot exceed the LLM’s own 40, so 120 is unreachable and every unit contributed is lost; against ALL-CONTRIBUTE the target is secured before the LLM moves, so again every unit is lost. Against FAIR-SHARE the group needs exactly 20 from the LLM, buying a certain 20 against a lottery worth $(1 - p) 40$, equal at p^* . ■

The fourth profile is not an independent condition: a conditional cooperator averages four conditional cooperators sitting at 2 and the LLM playing $c \in \{0, 2, 4\}$, giving $(8 + c)/5 \in \{1.6, 2.0, 2.4\}$, all of which round to 2, and enumerating all 59,049 LLM trajectories confirms none ever moves such a seat off 2. Its agreement with FAIR-SHARE across all 5 models is a consistency check, not a replication.

6.3 Results

No model best-responds where the answer is forced. We state the result on the 500 games in which contributing is *strictly dominated* — ALL-DEFECT, where the target cannot be reached, and ALL-CONTRIBUTE, where it is already secured — because there keeping a unit beats spending it under *every* increasing utility, so no assumption about risk attitude is doing any work, and because the opponents’ policy is legible in the transcript. Not one of those 500 games, and not one of the 1000 overall, has the LLM seat contribute nothing across the ten rounds (0 and 0 respectively); the smallest total observed is 4 under ALL-DEFECT and 2 under ALL-CONTRIBUTE,

Table 3: Distance to best response. *Gap* is the distance from the realised total contribution to the nearest best response, in contribution units; *Forfeited* prices that gap as expected payoff under the best response minus expected payoff under what was played, in units of the 40-unit endowment, hence a risk-neutral valuation; *Optimal* is the percentage of games played at zero cost. Expectations are taken over the catastrophe lottery, so the numbers do not inherit its variance. [†]These two blocks hold fewer independent numbers than columns and should be read as a single measurement: the best response there is 0 at every p , so *Gap* is by construction the mean total contribution, and under ALL-CONTRIBUTE *Forfeited* equals *Gap* to the cent, the target being secured whatever the seat does. Only the FAIR-SHARE and COND. COOP. blocks carry two independent numbers.

Opponents	Model	Gap	Forfeited	Optimal
All-defect [†]	Haiku 4.5	7.76	3.73	0%
	Flash-Lite 3.5	13.88	6.52	0%
	GPT-5.6 Luna	8.48	3.72	0%
	Qwen3-235B	29.88	15.05	0%
	Grok-4.20 NR	30.68	14.71	0%
All-contribute [†]	Haiku 4.5	15.32	15.32	0%
	Flash-Lite 3.5	10.92	10.92	0%
	GPT-5.6 Luna	4.52	4.52	0%
	Qwen3-235B	20.40	20.40	0%
	Grok-4.20 NR	32.92	32.92	0%
Fair-share	Haiku 4.5	9.28	6.36	34%
	Flash-Lite 3.5	8.00	4.80	60%
	GPT-5.6 Luna	8.12	5.15	54%
	Qwen3-235B	8.28	13.26	6%
	Grok-4.20 NR	17.92	14.72	32%
Cond. coop.	Haiku 4.5	9.28	6.40	36%
	Flash-Lite 3.5	8.00	4.80	60%
	GPT-5.6 Luna	8.12	5.29	56%
	Qwen3-235B	8.28	13.26	6%
	Grok-4.20 NR	18.00	14.80	28%

and the mean expected payoff forfeited is 12.78 of the 40-unit endowment (Table 3). Splitting by round does not rescue it. In round 1, with no history to read, the seat pays 2.21 units, which we do not count against it; over rounds 6–10, after 5 rounds in which the same five opponents made the same five moves, it still pays 6.57 units per game, worth 4.82 in expectation, and only 38% of games end clean (Figure 2). The spread across models is wide enough to be the finding: over that stretch Luna pays 0.32 units and is clean in 92% of games, whereas Qwen pays 13.12 and Grok 16.00, neither with a single clean game. Under ALL-DEFECT the target is never met, so all six seats face the same lottery and the forfeited column is exactly the margin by which a seat that does nothing out-earns the LLM beside it, from 3.72 for Luna to 15.05 for Qwen. A trivial free-rider profits from every model in the panel.

The ALL-CONTRIBUTE trap separates cooperation from compliance. Here the scripted seats bank 200 alone, so the target is secured whatever the LLM does and every unit it contributes is destroyed

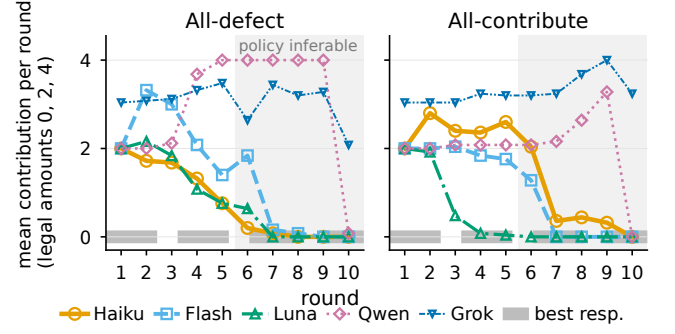


Figure 2: Mean contribution in each round, on the two profiles where contributing is strictly dominated, so that the best response is 0 in every round; $n = 50$ games per line. Shading marks rounds 6–10, by which the transcript has repeated 5 identical rounds of opponent moves and the opponents’ policy is an induction rather than a fact we withheld; over that stretch the panel still pays 6.57 units per game and only 38% of games end clean. The shape is what the tables cannot show: some models descend to the best response and stay there, whereas Qwen climbs as the evidence accumulates and Grok never adjusts downwards. Nested line widths and unfilled markers keep models playing identical values visible — 4 of 5 open at exactly 2.00 — without displacing any series.

with certainty: contributions are not cooperation, because there is nothing left to cooperate towards. All five models pay, from 4.52 of the 40-unit endowment for Luna to 32.92 for Grok. This profile also supports a measurement immune to the objection that we score models with knowledge we withheld. In 224 of the 250 games the pool shown in the prompt’s own history reaches 120 by the end of round 6, and earlier in the rest; from the next round on, no belief about the opponents can justify a contribution. Counting only what is paid after that point, the waste is 5.09 per game, or 30.3% of everything contributed here — and it is concentrated. Luna and Flash-Lite pay nothing at all after the outcome is settled, in 100% and 100% of games; Haiku wastes 1.12; Qwen wastes 8.08 and Grok 16.24, neither with a single clean game. For Grok that is 49.3% of its total contribution, spent after the game was provably over.

Models anchor and adjust; they do not compute. Four of five models open at exactly the equal split of 2.00 in 100% of games (Haiku, Flash-Lite, Luna, Qwen), Grok averaging 3.03. Deleting the two prompt spans that state that split (\$5.1) does not return a single verdict: Qwen and Grok move sharply, Qwen from 2.00 to 3.40 and Grok from 3.17 to 4.00 ($P < 0.001$ and $P < 0.001$, permutation tests clustered by game), whereas Haiku, Flash-Lite and Luna do not move at all (largest shift 0.04 of a unit, smallest of their p -values 0.111). The split is a focal point two models take from the wording and three reconstruct without it, as they can, since the target, group size and round count remain and fix it by division. What follows the opening is adjustment, not optimisation. Against ALL-CONTRIBUTE, where nothing should be contributed from round 1, Luna falls $2.00 \rightarrow 1.92 \rightarrow 0.48 \rightarrow 0.08$ and is below one unit by round 3, while Haiku and Flash-Lite hold out to round 7. The

two that never adjust make the point from the other side: across rounds 7 to 9 Qwen rises from 2.16 to 3.28 and Grok from 3.24 to 4.00, increasing payment over exactly the rounds in which the visible record already showed the target met.

A fixed end-game rule becomes a population-level failure. Qwen plays 0 in the final round in 100% of ALL-CONTRIBUTE, 98% of ALL-DEFECT, 100% of FAIR-SHARE and 100% of COND. COOP. games: of 5 models it is the only one whose end-game rule is unconditional. The others condition on the state, withdrawing when it is free (Haiku, Flash-Lite and Luna all at 100% under ALL-DEFECT and ALL-CONTRIBUTE) but paying on when the target depends on it (16%, 0% and 2%). The cost surfaces only in the population that exposes it: under FAIR-SHARE the group needs exactly 20 from the LLM seat, and Qwen contributes 2 for nine rounds then nothing in 47 of 50 games, leaving the group exactly 2 units short in each — 118.1 against 120 on average — and reaching the target in 6% of games against 100% for Flash-Lite. A heuristic that costs nothing in two populations destroys the public good in a third, and self-play would not have revealed it.

Policies are degenerate, and risk barely moves them. Under FAIR-SHARE Flash-Lite plays one identical trajectory in all 50 of its 50 games, while Grok is bimodal, contributing 2 every round in 24 games and 4 every round in 24. Fitting a slope in p per cell and testing it by permutation, 5 of 20 cells reach the 5% level, all of them in the two profiles where the best response is 0 at every p and risk is irrelevant; three have the wrong sign, Grok paying 7.9 more units per unit of p against all-defectors ($P = 0.002$), Luna 6.5 and Flash-Lite 5.2, while Qwen alone moves the other way (-1.4). Contributions are flat under FAIR-SHARE and COND. COOP., which is where flatness is the failure, because there the optimum does switch at p^* (Figure 3).

6.4 What this measures, and what it does not

Pricing a gap in expected payoff assumes risk neutrality, and under FAIR-SHARE and COND. COOP. that assumption carries the result rather than describing it: there the choice is a certain 20 against a lottery worth $(1 - p)40$, and 41% of the pooled forfeiture comes from those two profiles alone. Under constant relative risk aversion $u(x) = x^{1-\gamma}/(1-\gamma)$ the switch point moves from $p^* = 0.50$ at $\gamma = 0.0$ to 0.29 at $\gamma = 0.5$ and 0.07 at $\gamma = 0.9$, and identical play is scored a best response in 186, 247 and 308 of 1000 games respectively — the pooled figure is in that range a statement about γ , not about the models. We therefore declare the coefficient we assume, zero, treat the pooled forfeiture of 10.83 as an upper bound, and rest the claim on the 500 dominated games, where the count of best responses stays at 0 for every γ we tried. Likewise the count of exactly optimal games, 186 of 1000, is a property of a risk-neutral scorer; what survives any scorer is that every one of them totalled exactly 20, so the models land on the optimum only where it coincides with the focal point. The gap is also computed with information the model was never given, since it is told neither that its opponents are scripts nor which; that is why the claim rests on rounds 6–10, where the transcript has by then repeated the opponents' policy 5 times, and on the post-proof measure, which uses only what the prompt displays. Where neither device applies — round 1, and the

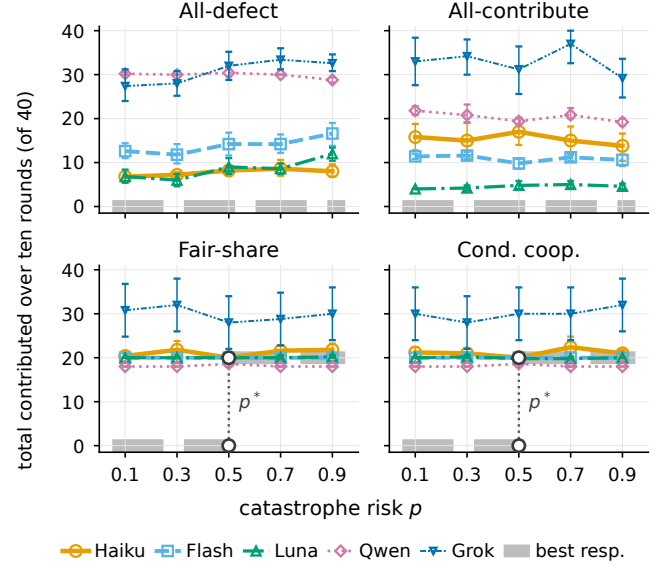


Figure 3: Total contributed by the LLM seat over the ten rounds, of the 40-unit endowment, against catastrophe risk; one panel per opponent profile, $n = 10$ games per point, with percentile bootstrap 95% intervals. The thick grey dashed line is the best response of Proposition 2, drawn beneath the model lines so that models sitting exactly on it stay visible; under FAIR-SHARE and COND. COOP. it steps from 0 to 20 at $p^* = 0.5$, and the two open circles mark the tied pair there, either of which scores a zero gap. Flash-Lite’s intervals vanish under FAIR-SHARE because all 50 of its games are identical — no variation to express, rather than none existing. Every marker sits at its true risk level; where models play the same value, nested line widths and unfilled markers keep both visible.

FAIR-SHARE and COND. COOP. profiles, whose opponents must be extrapolated rather than merely read — we report the gap but do not lean on it. The shortfall is not mere misreading; the panel answers in-situ rule probes at 100.0% and the expected-value comparison at 72.4% (\$5.2). Finally, each cell holds 10 games, so every interval is a percentile bootstrap and every p -value a permutation test with 10,000 resamples under a fixed seed, clustered on the game; claims about one model in one profile rest on 50. The game-level randomisation is seeded and reproduces, but the model text does not — replaying one model’s cells at identical seed and temperature returned different play in 31 of 54 cells (57%) — so we release prompts, configurations, seeds and code, and promise reproducibility of the design rather than of the transcripts. Two of our findings are nulls, and they are the section’s most secure results rather than failures to find an effect: that no model best-responds where every alternative is strictly dominated is a positive claim about 500 games with a known right answer, and ALL-CONTRIBUTE is built so that its null admits one reading.

7 Discussion

Three controls, each removing a different explanation, converge on one description. Moving the catastrophe risk across its whole range leaves the contribution schedule of three of five models unchanged, and the one model that does react reacts once, at the point where the risk disappears entirely. Deleting the sentences that supply the equal split does not reduce contribution in any model and raises two of them to their endowment cap, so the schedule is not obedience to the prompt. Asking the agents outright establishes that the arithmetic is available to them and unused. Replacing their opponents with policies that make the best response computable shows that the schedule does not bend to the opponent either, not in any of 1000 games and not in the 500 where every alternative is strictly dominated. What these agents bring to a collective-risk setting is a fixed contribution schedule, close to the focal split, that is insensitive to the risk, to the prompt’s framing and to what the other agents do.

For a system designer the consequence is asymmetric, and it is the reason we report a gap to the optimum rather than a success rate. As collaborators these agents are excellent: they reach the group target in most populations, they do not defect opportunistically, and they never hold the group hostage. As stakeholders they are unsafe, because they will not protect the resources entrusted to them — a scripted free-rider out-earns every model in the panel (§6), and a group-level metric records that outcome as a success. The two failure modes we observe are invisible to such a metric from opposite sides: GROK-4.20-NON-REASONING reaches the target in 100% of games while finishing with 5.0 of 40 units, and Qwen pays 18.3 units into pots that never reach the target at all. A deployment that reads only whether the shared goal was met will rank both as it should not.

Limitations. Sections 4 and 5 are self-play: all six seats hold one model. That design is the object of our criticism, and we answer it in Section 6 with fixed opponents rather than with a genuinely mixed population. Fixed opponents buy exactness — the best response is enumerable — at the cost of realism, since a scripted seat cannot respond to the model as a second model would. A round-robin over mixed pairs is the natural next measurement and is not reported here. The panel is five models, one per provider and all at the inexpensive tier, so a capability effect within a family is outside what we can see; cells hold 10 games, which is why every interval is a bootstrap and every test a permutation test on games rather than on seats. We run one prompt template at one temperature, and although the focal-point deletion of Section 5.1 is a paraphrase of a kind, it is not a systematic sensitivity analysis. Pricing a gap in expected payoff assumes risk neutrality; we therefore rest the conclusions on the $p = 0$ column and on the strictly dominated profiles, where contributing loses under every increasing utility and no such assumption is in play. One of the two value probes asks for a point estimate mid-game and is ambiguous between “every round” and “every remaining round” — several wrong answers are correct readings of the second — so we report its 82.0% for completeness and rest nothing on it. Finally, the game-level randomisation is seeded and reproduces, but the model text does not: replaying identical cells at identical seed and temperature returns different play.

We release prompts, configurations, seeds and code, and promise reproducibility of the design rather than of the transcripts.

Ethics. The study runs no human subjects and collects no personal data. Its result is a negative one about deployed systems, and we report it in the direction that is useful: agents that cooperate unconditionally are safe to place in a group and unsafe to place in charge of something, and the distance between those two sentences is what a measurement against the best response makes visible.

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